A MultiKernel Domain Adaptation Method for Unsupervised Transfer Learning on Cross-Source and Cross-Region Remote Sensing Data Classification

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***Abstract*— Labeling remote sensing data for classification is labor-intensive and time-consuming. Transfer learning (TL), under such context, is attracting increasing attention as it aims to harness information from data set of other regions where labels are readily available. The central topic of concern is to homogenize the large disparities of feature distribution of different data set through domain adaptation (DA). This article proposes a novel DA method for unsupervised TL, namely, multikernel jointly domain matching (MKJDM), which by definition considers multiple kernels as opposed to the currently popular single-kernel methods for measuring the distances between distributions. The single-kernel methods minimize the distances of feature distribution between the source domain (data set with training labels) and the target domain (data set to be classified) through, for example, maximum mean discrepancy (MMD) metric, formed under a kernel function mapping, while the multikernel version (MK-MMD) uses different kernel functions to encapsulate multiple aspects of distribution discrepancies, and is, therefore, more capable of distance minimization. Our MKJDM implementation also considers simultaneously aligning marginal and class conditional distributions and reweight for each instance, which further improves the performance. Two experiments performed on remote sensing images and multimodal data sets (i.e., Orthophoto and Digital Surface Models), with regions of different countries with distinctly different land patterns serving as source and target domain data, show that the overall accuracies are improved by 37.28% and 46.62% after applications of our MKJDM method. An additional comparative experiment with five state-of-the-art DA methods also demonstrates that our method achieves the best performance.**

***Index Terms*— Domain adaptation (DA), image classification, remote sensing, stable index, transfer learning (TL), unsupervised learning.**

I. Introduction

CLASSIFICATION for characterizing land cover on remote sensing images embraces a wide range of applications in many fields, such as land cover/land use change, urban growth, and disaster analysis [1], [2]. The often-used supervised classification methods require a large amount of labeled data for training, while in practical applications, such a labeled data set is either not readily available or expensive to acquire. A possible way to address such an issue is to harness information from data sets of other regions where labels are readily available (source domain). Several factors such as viewing angle of sensors, sun elevation angle differences, different kinds of sensors, atmospheric changes, seasonal variations affecting the phenology of vegetation, and land-cover patterns might cause a large difference between images acquired on different geographical areas or over the same area but at different times [3]. A nontransferred classifier, which is trained by the source data set and directly applied to the testing data set (target domain), may result in undesired classification results because of the large discrepancy between the source and target domains. Transfer learning (TL), which learns an adapted classifier for the target domain by utilizing the information from the existing available labeled data (source domain), has shown a promising prospect in computer vision field in recent years [3]–[5]. In our unsupervised TL task, labeled data for training are only available in the source domain and the source and target domain are different in the data distribution. In the TL field, domain adaptation (DA) is used for such a TL task [6]. DA aims to transform the feature representations of both domains, such that the joint probability distributions of the source domain and the target domain become more similar, thus leading to smaller discrepancies between the source and target domains. Through DA, TL can greatly improve performance and avoid expensive data labeling efforts at the same time.

The distance measurement method between the source and target domains is important for DA. Most existing DA methods use maximum mean discrepancy (MMD) as the distance measurement for the statistical distance of distributions in a projected space [6]–[9]. Reproducing Kernel Hilbert Space (RKHS) is a widely used projected space which corresponds to a kernel. After choosing the specific kernel of the RKHS'Ca linear mapping matrix, which is able to project the source and target data sets into a third space (RKHS), can be computed by minimizing MMD. The transferred source and target domains are expected to have similar feature distributions. The current DA methods based on single-kernel MMD find that

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Gaussian kernel is the most effective, while the bandwidth of kernel is often kept as a constant and this might not be optimal to account for a wide range of feature distributions presented in the source and target domains [7], [9]–[15]. To overcome the drawback of the single kernel and achieve a better learning capability, multiple kernel learning (MKL) has been proposed to use a combination of a series of base kernels to replace the single kernel [19]. The reason that MKL has an advantage over single-kernel learning is that different kernels are able to extract information from different sources (representations or feature subsets) [13]–[15]. For example, a combination of Gaussian kernels with different bandwidths can be seen as a combination of multiscale kernels to address classification problems containing objects varying significantly in scales [18]. Generally, MKL aims to find an optimal linear or nonlinear combination of a series of base kernels [19]. Since nonlinear combinations usually lead to nonconvex optimization problems with very high computational complexities and also suboptimal performances [13], [20], linear multiple kernel combinations are most used in the field of remote sensing. Inspired by the idea of MKL, we consider computing MMD in an RKHS associated with optimal linear kernel combination. Multikernel MMD (MK-MMD) has been proposed as an optimal linear kernel combination method for MMD (see [21, Sec. 2.4] for details). MK-MMD is able to determine an optimal kernel formed by a weighted combination of multiple kernels based on the source and target data sets. Based on MK-MMD, we propose a novel DA method, namely, multi-kernel jointly domain matching (MKJDM), for TL on remote sensing data. Our proposed MKJDM minimizes the discrepancy between the source and target domains through: 1) simultaneously aligning the marginal distributions (class-free feature distribution) and conditional distributions (per-class feature distribution) and 2) adaptively reweighting learning samples based on their contributions in DA, to further improve the classification accuracy.

Our experiments are performed both on very high resolution (VHR) multispectral data sets and multimodal remote sensing data sets [including pixel-wise overlaid Orthophoto and Digital Surface Models (DSM)] with different scene contents. The rest of the sections are organized as follows: Section II introduces the related works in DA, and basics of the MK-MMD and MMD measures, as well as two methods mostly relevant to our work: 1) transfer component analysis (TCA) and 2) transfer joint matching (TJM). Section III describes the proposed MKJDM method in detail, and Section IV presents the experimental data set and feature extraction methods. The experimental results and the analysis are performed in Section V. Section VI concludes this article by analyzing the pros and cons of the proposed method.

II. Related Work

In the field of computer vision, DA methods have been widely used in the cases when training labels are not available and have been applied to unsupervised learning and classification problems [12], [22]–[24]. The DA methods are often performed by minimizing the distance between the feature distributions of source and target domains. The distances can be defined using different approaches, for example, CORrelation ALignment (CORAL) [23] utilized the second-order statistics (covariance) of the source and target features as the distance between the two domains; Geodesic Flow Kernel (GFK) embedded source and target data sets in a Grassmann manifold and regarded the distance between two domains in such a manifold as the distance metrics [25]. MMD, among these metrics, is the most widely for DA [6], [9], [26]. TCA, for example, is a classic DA method based on this metric which optimizes the summed MMD distances of all instances between the source and target domain. It has shown to be capable of generating new feature representations for both source and target domain with smaller distances. Applications of TCA using remote sensing data sets for TL have demonstrated that this method is able to improve the accuracy of cross-scene classification to a notable level [27]. To further improve the performance of TCA, Long *et al.* [12] proposed a TJM method that incorporates an instance reweighting method into the optimization framework of TCA to reduce the effect of some training samples which are irrelevant with the same-class samples in the testing data set. This type of methods takes the marginal distribution alignment as the goal for optimization, while this may perform well only when their feature domains have similar conditional distributions: for example, if the source data set is from a residential area and the target data set is from a downtown area, the proportion of buildings and vegetation is significantly different. Merely aligning marginal distribution might make the transferred classifier recognize some buildings as vegetation. In the cases where the conditional distributions of the source and target domains are significantly different, minimizing their joint probability distributions distances can be particularly useful. By taking advantage of the pseudo target label generated by a nontransferred classifier, the conditional distributions of the feature domains can be built. With this idea, the joint distribution adaptation (JDA) method proposed by Long *et al.* [28] incorporated conditional distributions with the marginal distributions into the distance minimizing process and reported a better accuracy than aligning the marginal distributions alone.

MMD can be normally well estimated for two feature distributions mapped into an RKHS, while how to choose an appropriate RKHS is a key issue to success. The existing DA methods fix an RKHS for a particular task, but adaptively choosing optimal kernels can be crucial to further improve the DA accuracy. Gretton *et al.* [21] proposed MK-MMD for kernel choice which minimizes Type II error (the probability of two samples belonging to different distributions identified as the same), given an upper bound on Type I error (the probability of two samples belonging to the same distribution identified as). In [21], a family of kernels is linearly combined by different weights which can be computed adaptively by the samples in the source and target domains. This makes it possible to apply the optimal RKHS kernel in DA. Based on the MK-MMD measurement, our method is built on two basic probability alignment methods: 1) TCA and 2) TJM, and these two methods are respectively introduced in the Appendix.

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In this section, we first introduce the widely used and basic MMD measurement (see Section II-A), which is also the basic version of the more advanced MK-MMD (MK-MMD, introduced in Section II-B) used in our proposed work.

A. MMD

MMD is accurate in finding samples that were generated from the same distribution [21]. Therefore, MMD can be seen as a measurement for the distance between probability distributions based on RKHS [29]. Let $\mathbf{X}_S \in \{\mathbf{x}_{S_i}\}_{i=1}^{n_s}$ and $\mathbf{X}_T \in \{\mathbf{x}_{T_j}\}_{j=1}^{n_t}$ be the feature sets of the source domain Δ_s and the target domain Δ_t over all the classes, respectively, where n_s is the number of samples in the source domain and n_t is the number of samples in the target domain; the empirical estimate of the distance between $\mathbf{X}_S$ and $\mathbf{X}_T$ can be defined by MMD as

$$\text{Dist}\,(\mathbf{X}_S, \mathbf{X}_T) = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi(\mathbf{x}_{S_i}) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi(\mathbf{x}_{T_j}) \right\|_2 \tag{1}$$

where ϕ is a nonlinear transformation that maps the feature vector to a universal RKHS [32]. Instead of using an explicit function ϕ, Pan *et al.* [30] proposed to reformulate this as a kernel learning problem, in which the MMD distance between $\mathbf{X}_S$ and $\mathbf{X}_T$ can be written as [31]

$$\text{Dist}\,(\mathbf{X}_S, \mathbf{X}_T) = \text{tr}(\mathbf{KM}) \tag{2}$$

where $\mathbf{K} = \phi([\mathbf{X}_S, \mathbf{X}_T])^{\mathrm{T}}\phi([\mathbf{X}_S, \mathbf{X}_T]) \in \mathbf{R}^{(n_s+n_t)\times(n_s+n_t)}$ is the kernel matrix in RKHS and $\mathbf{M} \in \mathbf{R}^{(n_s+n_t)\times(n_s+n_t)}$, with $M_{ij} = 1/n_s^2$, if $\mathbf{x}_i, \mathbf{x}_j \in \Delta_s$; $M_{ij} = 1/n_t^2$ if $\mathbf{x}_i, \mathbf{x}_j \in \Delta_t$; otherwise, $M_{ij} = (-1/n_s n_t)$. The readers may refer [31] for more details on this formulation.

B. MK-MMD

Let H_u be the RKHS endowed with a Gaussian Radial Basis Function (RBF) kernel K_u whose bandwidth is σ_u. ϕ_u is the mapping function which can map the original data into H_u. Combining a family of d kernels, the MK-MMD can be defined as

$$\begin{aligned} H_{\text{MK}} &= \text{Dist}(\Delta_s, \Delta_t) \\ &= \sum_{u=1}^{d}\beta_u \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi_u(\mathbf{x}_{S_i}) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi_u(\mathbf{x}_{T_j}) \right\|_2 \end{aligned} \tag{3}$$

where β_u is the weight of the uth kernel k_u, $\sum_{u=1}^{d}\beta_u = 1(\beta_u \geq 0)$. We denote $\beta = (\beta_1, \beta_2, \ldots, \beta_d)^{\mathrm{T}} \in \mathbf{R}^{d\times 1}$ and $\eta = (\eta_1, \eta_2, \ldots, \eta_d)^{\mathrm{T}} \in \mathbf{R}^{d\times 1}$ each η_u is the MMD via kernel k_u. With this notation

$$\eta_{\text{MK}} = \beta^{\mathrm{T}}\eta. \tag{4}$$

Following [21], we seek to learn the optimal kernel parameters β for MK-MMD by jointly maximizing the test power and minimizing the Type II error. Then, the multikernel problem becomes equivalent to a convex quadratic program with a unique solution, given by

$$\begin{aligned} &\min \beta^{\mathrm{T}}(\mathbf{Q} + \lambda\mathbf{I})\beta \\ &\text{s.t.} \sum_{u=1}^{d}\beta_u = 1 \quad (\beta_u \geq 0) \end{aligned} \tag{5}$$

where $\lambda = 10^{-3}$ is a small regularization parameter to make the problem well-defined and $\mathbf{Q} \in \mathbf{R}^{d\times d}$ is the covariance matrix of $\mathbf{L} \in \mathbf{R}^{d\times(n_s/2)}$. $\mathbf{L}$ can be computed as follows:

$$\begin{aligned} \mathbf{L}(u,i) &= k_u(\mathbf{x}_{S_{2i-1}}, \mathbf{x}_{S_{2i}}) + k_u(\mathbf{x}_{T_{2i-1}}, \mathbf{x}_{T_{2i}}) \\ &\quad - k_u(\mathbf{x}_{S_{2i-1}}, \mathbf{x}_{T_{2i}}) - k_u(\mathbf{x}_{S_{2i}}, \mathbf{x}_{T_{2i-1}}) \end{aligned} \tag{6}$$

$$k_u = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{2\sigma_u^2}\right). \tag{7}$$

Our base kernel set $\{k_u\}_{u=1}^{d}$ contains d univariate kernels with fixed bandwidth. After solving the quadratic program in (5), the optimal linear combination for MK-MMD can be acquired. The readers may refer [21] for more details about MK-MMD.

III. Proposed MKJDM

With the introduction of relevant works and basics, in this section, a DA method that builds on TJM using MK-MMD is introduced, but with three main improvements to further improve the DA accuracy.

1) TJM [12] merely aligns marginal distribution. Our MKJDM method simultaneously aligns the marginal and conditional distributions in a single optimization step.
2) We use MK-MMD to measure the distance of the source and target domains, which can accommodate different aspects of distribution discrepancies.
3) Because the reweighting instance method in MKJDM only reduces the weights of instances by row-sparsity, MKJDM only reweights the instances belonging to the classes appear in the classification result. Utilizing contextual information generated by pseudo labels is assumed to be more robust for conditional probability alignment.
4) We apply the TCA result as the pseudo label set for estimate conditional distribution of the target domain.

To consider conditional distribution and marginal distribution at the same time, the MK-MMD of two domains in H_u can be expressed as

$$\begin{aligned} \eta_u \approx &\left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi_u(\mathbf{x}_{S_i}) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi_u(\mathbf{x}_{T_j}) \right\|_{H_u}^2 \\ &+ \sum_{c=1}^{C}\left\| \frac{1}{n_c}\sum_{\mathbf{x}_{S_i}\in\Delta_s^{(c)}}\phi_u(\mathbf{x}_{S_i}) - \frac{1}{m_c}\sum_{\mathbf{x}_{T_j}\in\Delta_t^{(c)}}\phi(\mathbf{x}_{T_j}) \right\|_{H_u}^2 \end{aligned} \tag{8}$$

where $c \in \{1, 2, \ldots, C\}$ is the distinct class label, and n_c and m_c are the number of instances belonging to class c in the source and target domains, respectively. Similar to TJM and

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JDA, the distance between the empirical means of the two domains in (8) can be written as

$$\text{Dist}\,(\Delta_s, \Delta_t) = \text{tr}\left(\mathbf{K}_u \sum_{c=0}^{C} \mathbf{M}_c\right) \tag{9}$$

where $\mathbf{M}_0$ is corresponding to the marginal distribution distance measured by MK-MMD, with $\mathbf{M}_0(i, j) = 1/n_s^2$, if $\mathbf{x}_i, \mathbf{x}_j \in \Delta_s$; $\mathbf{M}_0(i, j) = 1/n_t^2$ if $\mathbf{x}_i, \mathbf{x}_j \in \Delta_t$; otherwise, $\mathbf{M}_0(i, j) = -1/n_s n_t$ and the MK-MMD conditional distance matrix $\mathbf{M}_c$ $(1 < c < C)$ can be computed as (10), $\mathbf{K}_u \in \mathbf{R}^{(n_s+n_t)\times(n_s+n_t)}$ is the kernel matrix of H_u, $y(\mathbf{x}_i)$ is the label of $\mathbf{x}_i$, $\Delta_s^{(c)} = \mathbf{x}_i : \mathbf{x}_i \in \Delta_s \wedge y(\mathbf{x}_i) = c$, $\hat{y}(\mathbf{x}_j)$ is the pseudo label of $\mathbf{x}_j$, $\Delta_t^{(c)} = \mathbf{x}_j : \mathbf{x}_j \in \Delta_t \wedge \hat{y}(\mathbf{x}_j) = c$. In MKJDM, the pseudo label set is generated by a classifier transferred by TCA. Because TCA is independent of any label information, such a method is a benefit for improving the accuracy of the pseudo label set. Readers may find more information about TCA in the Appendix

$$(M_c)_{ij} = \begin{cases} \dfrac{1}{n_s^{(c)} n_s^{(c)}}, & \mathbf{x}_i, \mathbf{x}_j \in \Delta_s^{(c)} \\ \dfrac{1}{n_t^{(c)} n_t^{(c)}}, & \mathbf{x}_i, \mathbf{x}_j \in \Delta_t^{(c)} \\ \dfrac{-1}{n_s^{(c)} n_t^{(c)}}, & \mathbf{x}_i \in \Delta_t^{(c)}, \mathbf{x}_j \in \Delta_s^{(c)} \text{ or} \\ & \mathbf{x}_i \in \Delta_s^{(c)}, \mathbf{x}_j \in \Delta_t^{(c)} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

In this article, we employ the Gaussian RBF kernel

$$\mathbf{K}_u(i, j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma_u^2}\right) \tag{11}$$

where σ_u is the kernel bandwidth. To acquire a family of kernels, assume σ_u varies from 0.025 to 2 with a step-size of 0.025. And then, the MK-MMD of the mapped two data sets can be written as

$$\eta_{MK} = \text{tr}\left(\mathbf{W}^{\mathrm{T}} \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right) \tag{12}$$

where $\mathbf{K}_M = \sum_{u=1}^{d} \beta_u \mathbf{K}_u$ is the combined multikernel and β_u can be computed by solving (5). Similar to TJM, we impose the $\ell_{2,1}$-norm structured sparsity regularization on the transformation matrix for the source domain $\mathbf{W}_s$ to control the complexity of the transformation matrix $\mathbf{W}$ and reweight the instances belonging to certain classes in the source domain. The optimization can be formalized as

$$\begin{aligned} &\min\ \text{tr}\left(\mathbf{W}^{\mathrm{T}} \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right) + \lambda\left(\|\mathbf{W}_s\|_{2,1} + \|\mathbf{W}_t\|_F^2\right) \\ &\text{s.t. } \mathbf{W}^{\mathrm{T}} \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}} \mathbf{W} = \mathbf{I}. \end{aligned} \tag{13}$$

According to the constrained optimization theory, we denote $\mathbf{Z} = \text{diag}(z_1, z_2, \ldots, z_k) \in \mathbf{R}^{k\times k}$ as the Lagrange multiplier and derive the Lagrange function for (13) as

$$\begin{aligned} &\text{tr}\left(\mathbf{W}^{\mathrm{T}} \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right) \\ &\quad + \text{tr}\left(\left(\mathbf{I} - \mathbf{W}^{\mathrm{T}} \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right) \mathbf{Z}\right) \\ &\quad + \lambda(\|\mathbf{W}_s\|_{2,1} + \|\mathbf{W}_t\|_F^2). \end{aligned} \tag{14}$$

Setting the derivatives of (14) with respect to $\mathbf{W}$ to zero yields

$$\left(\mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} + \lambda \mathbf{G}\right) \mathbf{W} = \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}} \mathbf{W} \mathbf{Z}. \tag{15}$$

$\|\mathbf{W}_s\|_{2,1}$ is a nonsmooth function at zero, thus we compute its sub-gradient as $(\partial(\|\mathbf{W}_s\|_{2,1} + \|\mathbf{W}_t\|_F^2))/(\partial \mathbf{W}) = 2\mathbf{G}\mathbf{W}$, where $\mathbf{G}$ is a diagonal sub-gradient matrix [12]

$$\mathbf{G}(i, i) = \begin{cases} \dfrac{1}{2\,\|\mathbf{W}(i, :)\|_1}, & \mathbf{x}_i \in \Delta_s^{\mathbf{c}}, \|\mathbf{W}(i, :)\|_1 \neq 0 \\ 0, & \mathbf{x}_i \in \Delta_s^{\mathbf{c}}, \|\mathbf{W}(i, :)\|_1 = 0 \\ 1, & \text{otherwise} \end{cases} \tag{16}$$

where $\Delta_s^{\mathbf{c}}$ is the source domain's instances belonging to the classes appear in the pseudo labels acquired by a nontransferred classifier. Multiplying both sides in formula (15) on the left by $\mathbf{W}^{\mathrm{T}}$, we obtain

$$\mathbf{W}^{\mathrm{T}} \left(\mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}} + \lambda \mathbf{G}\right) \mathbf{W} = \mathbf{W}^{\mathrm{T}} \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}} \mathbf{W} \mathbf{Z}. \tag{17}$$

Substituting (23) into (20), the optimization becomes

$$\min\ \text{tr}\left(\left(\mathbf{W}^{\mathrm{T}} \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right)^{-1} \mathbf{W}^{\mathrm{T}} \left(\lambda \mathbf{G} + \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}}\right) \mathbf{W}\right) \tag{18}$$

or

$$\max\ \text{tr}\left(\left(\mathbf{W}^{\mathrm{T}} \left(\lambda \mathbf{G} + \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}}\right) \mathbf{W}\right)^{-1} \mathbf{W}^{\mathrm{T}} \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}} \mathbf{W}\right). \tag{19}$$

In kernel Fisher discriminant (KFD) [32], the solution of $\boldsymbol{\alpha}$ for maximizing the $(\boldsymbol{\alpha}^{\mathrm{T}} \mathbf{M} \alpha)/(\boldsymbol{\alpha}^{\mathrm{T}} \mathbf{N} \alpha)$ is the leading eigenvectors of $\mathbf{N}^{-1}\mathbf{M}$. Therefore, the transformation matrix $\mathbf{W} \in \mathbf{R}^{(n_s+n_t)\times D}$ in (19) is the eigenvectors corresponding to the D leading eigenvalues of $(\lambda \mathbf{G} + \mathbf{K}_M \sum_{c=0}^{C} \mathbf{M}_c \mathbf{K}_M^{\mathrm{T}})^{-1} \mathbf{K}_M \mathbf{H} \mathbf{K}_M^{\mathrm{T}}$, where D is the dimensions of the data. By multiplying $\mathbf{W}^{\mathrm{T}}$ by the combination kernel matrix $\mathbf{K}_M$ on the right side, the original domains can be mapped into multikernel RKHS. Now, the new feature set $\mathbf{D}_{\text{new}}$ is acquired

$$\mathbf{D}_{\text{new}} = \mathbf{W}^{\mathrm{T}} \mathbf{K}_M. \tag{20}$$

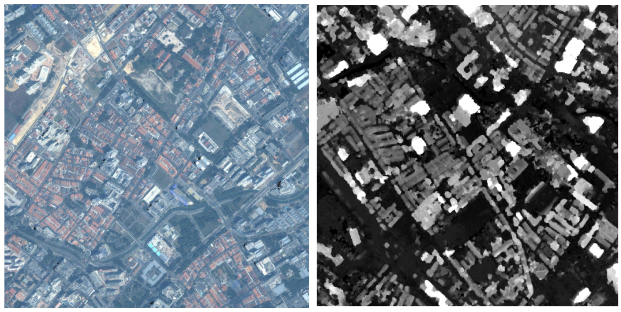
The first n_s columns of $\mathbf{D}_{\text{new}}$ is the new (transferred) feature representation in source domain and the last n_t columns of $\mathbf{D}_{\text{new}}$ is the new feature representation in target domain. And then we can classify the testing image by a simple statistical classifier using the new features.

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(a) (b)

Fig. 1. San Fernando, Argentina (the source domain). (a) Orthophoto. (b) DSM.



(a) (b)

Fig. 2. Rochor, Singapore (the target domain in the first experiment). (a) Orthophoto. (b) DSM.



(a) (b)

Fig. 3. Port-au-Prince, Haiti (the target domain in the second experiment). (a) Orthophoto. (b) DSM.

IV. Experiment Data and Processing

A. Experiment Data Set

In this article, we used three urban areas from different continents with different land covers. Our first study area is a 3.5-km^2 urban area in San Fernando (latitude 34° 27' S and longitude of 58° 37' W) which is a city in the Gran Buenos Aires, in Argentina, and capital of the San Fernando Partido [see Fig. 1(a)]. The second study area (see Fig. 2) is Rochor which is a region from Singapore (latitude 1° 17' N and longitude of 103° 50' E) and the third study area (see Fig. 3) is Port-au-Prince, Haiti (at latitude 18° 32' N and longitude of 72° 20' W), respectively.

In this article, we use the Argentina data set as the source domain and use the other two data sets as the target domains.

B. Data Description and Processing

The experiments are conducted on three VHR images (with or without DSM) obtained from Argentina, Singapore, and Haiti, respectively. The main reason involving DSM is that we would like to understand how our method works on multimodality data, which in this case, "Image + DSM," can be a fair common and standard as, for example, photogrammetry products. DSM also brings additional benefits as it is able to provide height information which is relatively stable across different domains. By using DSM features, the feature transferability is better so that we can generate an acceptable classification result after MKJDM. The Argentina data set is a VHR orthophoto, and DSM over San Fernando with a size of 6000 × 6000 pixels is computed by a hierarchical Semi-Global Matching algorithm (Hirschmuller *et al.* [35]) through RSP software [33]–[35]) using source satellite data from John's Hopkins University Applied Physics Lab (JHUAPL) [36], [37] with the worldview-3 satellite which provides eight spectral bands (red, red edge, coastal, blue, green, yellow, near infrared-1, and near infrared-2), with a 0.31-m panchromatic resolution. The DSM of this area is shown in Fig. 2(b).

The Singapore data set is an IKONOS Orthophoto [with DSM shown in Fig. 3(b)] [38] which contains four bands (blue, green, red, and near-infrared). It was obtained over the Singapore Rochor area with 1500 × 1500 pixels in 2002 with 1 m resolution. And the Haiti data set (Orthophoto + DSM) is obtained by the Geoeye-1 satellite over Port-au-Prince, the capital of Haiti. It was acquired in 2012 with 1.84 m spatial resolution and contains four bands (blue, green, red, and near-infrared) with 2001 × 2001 pixels. We can see that there are large discrepancies between the source domain (Argentina data set) and the target domains (Singapore data set and Haiti data set), such as sensor, resolution, the number of bands, observation area, and observation time. It is difficult for a classifier trained by the source domain to classify the images in target domains directly. However, our DA method is able to address this issue. Before implementing our idea, we need to pre-process the data sets to extract the original feature sets used to classify. In the remainder of this section, we explain how to extract the feature sets used in this article.

By processing the DSM with the morphology reconstruction algorithm, we can acquire the DSM features which make use of top-hat by reconstruction (THR) and top-hat by erosion (THE) with different scales [39]. Let the normalized DSM image be J, and J can be reconstructed as $B_{J,I}$ from a marker by finding the maximum of I, which is marked by J. I is derived from an erosion operation from J by a structuring element e. Considering two types of morphological top-hats, THR and THE can be simply defined as follows:

$$\mathrm{THR}(J, e) = J - B_{J,I} \tag{21}$$

$$\mathrm{THE}(J, e) = J - I. \tag{22}$$

These two top-hats can compensate for each other. As the sizes of the urban objects vary a lot, we use a series of structuring elements $\{e_i\}_{i=1,2,\ldots,N}$, respectively, to construct the multi-scale dual morphological top-hat profile (DMTHP)

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features

$$\mathrm{DMTHP}(J)_N = \{\mathrm{THR}(J, e_1), \mathrm{THR}(J, e_2), \ldots, \mathrm{THR}(J, e_N), \mathrm{THE}(J, e_1), \mathrm{THE}(J, e_2), \ldots, \mathrm{THE}(J, e_N)\}. \tag{23}$$

In this article, we use THR and THE with five different scales: $\{e_i\}_{i=1,2,\ldots,N} = \{10, 30, 70, 120, 210\}$. The readers may refer [40] for more information regarding DMTHP. Since the MMD distance matrix $\mathbf{M} \in \mathbf{R}^{(n_s+n_t)\times(n_s+n_t)}$, MMD-based DA algorithms need too much memory when the number of samples is large. Considering the limit of memory, we use an object-based classification method where a mean shift method is used for segmentation. The 1400 training samples are evenly selected from the source domain, including 473 building objects, 141 road objects, 520 tree objects, 48 barren objects, 83 grass objects, 42 ground objects, and 93 water objects. We repeated the road, barren, grass, ground, and water objects to balance the training data set. Finally, we got a balanced labeled data set containing about 500 samples for each class. For accuracy verification purposes, we have also labeled the target domain data sets. The feature vectors used in this article contain four spectral features (red, green, blue, and near-infrared1) and ten DMTHP features. The other four bands in the Worldview-3 image are discarded in order to ensure these three data sets have the same feature dimensions. All of the features are normalized to the range of [0, 1], and a random forest classifier is used to conduct all the experiments.

V. Results and Discussion

We compare MKJDM with some state-of-the-art DA methods—TCA [7], JDA [11], GFK [25], CORAL [23], and TJM [12]—in the experiments. We test MKJDM on both multimodel data sets (Orthophoto and DSM) and multi-spectral data sets (only have orthophoto). In addition, to understand the role of the MK-MMD metric, we have implemented a single-kernel version of the MKJDM, denoted as JDM for comparison. The results show that our approach is able to improve classification accuracy in almost all cases (with/without DSM information). The statistical analysis of our algorithm in this article uses multi-modal data as it achieves reasonably better improvements. Overall accuracy (OA) and Kappa index of agreement (KIA) [41] are used as metrics for accuracy evaluation.

A. Experiment 1: Singapore Data Set as the Target Domain Data

In this experiment, seven classes of interest have been interpreted, including buildings, roads, trees, bare land, grasses, ground, and water. The DSM of Singapore data set is shown in Fig. 2(b) and the ideal classification result (from target to target: taking the labeled target samples to train the classifier, and note these were not used in the DA) is shown in Fig. 4.

From Figs. 1 and 2, we observe that there are many different aspects of the two scenarios such as the size of a typical building, the spectral reflection of the roofs, and the width of the river. First, we adopt a family of RBF kernels whose bandwidths range between [0.025, 2] with a step-size of 0.025. Then, the final classification result acquired by MKJDM is shown in Fig. 5(b). As a comparison, the classification result acquired by a nontransferred RF classifier is shown in Fig. 5(a).

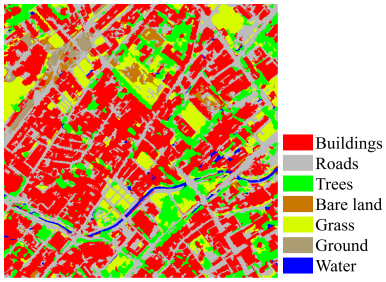


Fig. 4. Corresponding supervised classification result (from target to target).

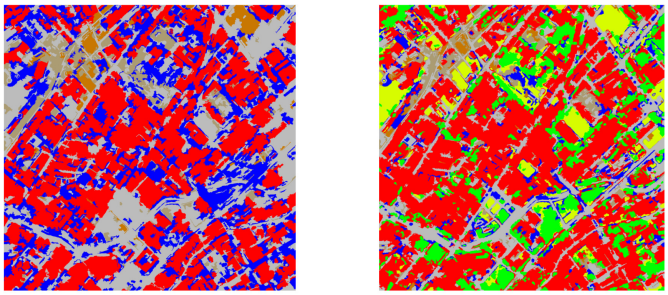


Fig. 5. (a) Result acquired by a nontransferred RF classifier (OA = 37.30% and KIA = 24.94%). (b) Result acquired by an RF classifier transferred by MKJDM (OA = 74.58% and KIA = 62.58%).

TABLE I
Results of Different DA Methods for Experiment 1

DA Method	Overall Accuracy (OA)	Kappa Index (KIA)
Src-Tar (with DSM)	0.3730	0.2494
Tar-Tar (with DSM)	0.8709	0.8196
TCA (with DSM)	0.5373	0.4328
JDA (with DSM)	0.6633	0.5536
GFK (with DSM)	0.3194	0.1655
CORAL (with DSM)	0.3703	0.2437
TJM (with DSM)	0.5388	0.4325
JDM (with DSM)	0.7031	0.6041
MKJDM (with DSM)	**0.7458**	**0.6258**
Src-Tar (image only)	0.3274	0.1977
Tar-Tar (image only)	0.8280	0.7574
MKJDM (image only)	0.5578	0.3936

In Fig. 5(a), there are only five classes (buildings, waters, bare land, ground, and road) in the classification result. Therefore, in MKJDM, we only reweight the instances belonging to these five classes. From the comparison in Fig. 5, we can see that MKJDM is able to improve the classification performance greatly (the OA is improved from 37.30% to 74.58% and the KIA is improved from 24.94% to 62.58%).

In order to further verify the effectiveness of MKJDM, some state-of-the-art DA methods are compared with MKJDM in Table I. Src-Tar represents the result acquired by a nontransferred classifier (classifier trained using the source labeled data

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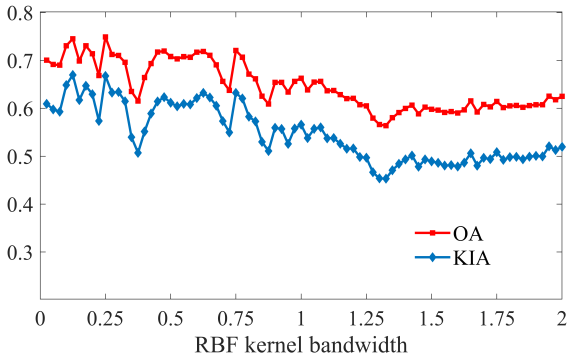


Fig. 6. Results acquired by JDM (single-kernel version of MKJDM) with different kernel bandwidths.

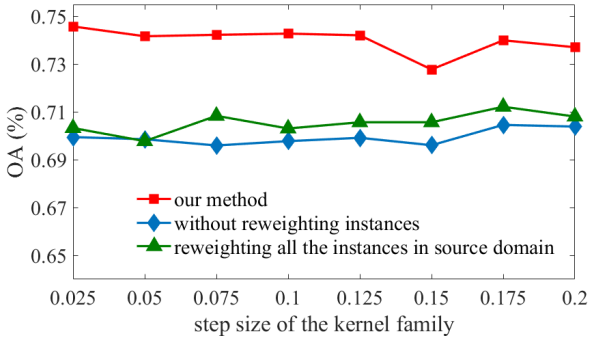


Fig. 7. Results acquired by reweighting all the instances, no instances, and the instances belonging to certain classes in the source domain.

and applied directly to the target data set). Tar-Tar represents the supervised method, and the result is acquired by fivefold cross-validation. Note that the JDM method is also proposed by us in this article. Its result is used to verify the validity of the multi-kernel method.

We can see from Table I that almost all the DA methods are able to improve the classification effect. The OA of MKJDM is 37.28% higher than Src-Tar and 8% higher than the second-best DA method. The comparison between MKJDM and JDM shows that the use of MKMMD is able to improve the accuracy by more than 4%. The KIA of MKJDM is 7.22% higher than the second-best DA method. In the case without using DSM features, the improvement from non-TL (OA: 32.47%) to our proposed method (OA: 55.78%) is also significant. In a single kernel case, the bandwidth parameter associated with the RBF used in our method is 0.5.

Fig. 6 shows the performance varies with different kernel bandwidth. By using multi-kernel measurement (MK-MMD), the classification performance is much less sensitive as the reprojection to RKHS can be to appropriate kernels. In addition, MKJDM also reweights some instances in the source domain. We fix the range of kernel bandwidth as [0, 2] and change bandwidth step size in the kernel family. With the change of step-size, the number of kernels in MKJDM also changes. Under different kernel family, we compare the results of MKJDM in three cases: 1) MKJDM with reweighting (our method); 2) MKJDM (without reweighting any instances); and 3) MKJDM (reweighting all the instances in source domain). The results are shown in Fig. 9.

In Fig. 7, we can see reweighting of instances improves the performance stably and reweighting part of samples are also better (about 3% higher in term of OA) than reweighting of all the instances in the source domain. From Fig. 8, we can see the step-size of bandwidths in the kernel family is not sensitive for MKJDM. Table II shows the comparison of class-wise F1-score between Src-Tar, Tar-Tar, JDM, MKJDM, and MKJDM without reweighting instance. Since MKJDM simultaneously considers marginal and conditional distributions, MKJDM is able to improve the F1-score in all the classes as shown in Table II. The reweighing strategy and MK-MMD are both able to further improve the DA performance. In some classes, MKJDM achieves a similar F1-score to Tar-Tar.

TABLE II
COMPARISON OF CLASS-WISE F1-SCORE FOR EXPERIMENT 1

Class	Src-Tar	Tar-Tar	JDM	MKJDM (without reweighting)	MKJDM
Buildings	0.7798	0.9203	0.8658	0.8604	0.8802
Roads	0.4971	0.8168	0.7772	0.7615	0.7395
Trees	0.0000	0.8727	0.7886	0.7707	0.8498
Bare land	0.0215	0.7884	0.0000	0.4361	0.5419
Grass	0.0000	0.8409	0.6441	0.6654	0.8382
Ground	0.2166	0.5811	0.3292	0.2417	0.3399
Water	0.0186	0.6325	0.0008	0.0000	0.0573

Using MKJDM, the F1-scores of some classes (e.g., buildings and trees) are improved to over 0.8. However, the F1-scores of some classes (e.g., bare land and water) are still very low. Because the per-class difference between the source and target domains may vary. For example, the features of tree class between source and target domains can be more similar than the bare land area, evidenced by the fact that Src-Tar F1-score for trees is much higher than the bare land. In general, classes with less difference are easier to be aligned as the high feature space vicinity allows good linear approximation which is used by the alignment algorithm.

Fig. 8 shows the comparison of class-wise original data distributions and class-wise transferred data (after MKJDM) distributions.

Considering joint distributions of the source and target domains, MKJDM maps the two data sets into Hilbert space associated with the combined kernel. After MKJDM, all the class data distributions are dramatically closer. It is the fundamental reason why MKJDM can effectively improve the classification accuracy.

B. Experiment 2: Haiti Data Set as the Target Domain Data

In this data set, six classes were interpreted as Experiment 1. The data set is shown in Fig. 4(a), and the ideal classification result (from target to target) is shown in Fig. 9. In this experiment, six classes of interest have been interpreted, including buildings, roads, trees, bare land, grasses, and ground. The DSM of the Haiti data set is shown in Fig. 4(b).

By visual comparison, there is a large discrepancy between the source (Argentina, southern hemisphere) and the target

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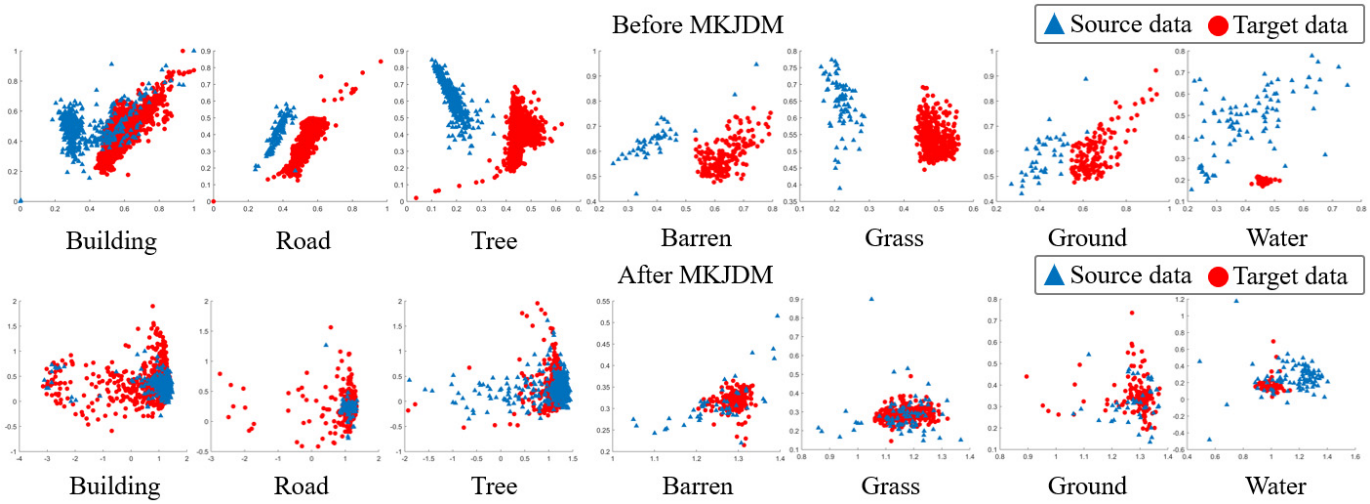


Fig. 8. Comparison of original data distributions and transferred data distributions (The 2-D coordinates in the figure are the first two features of the data before and after DA).

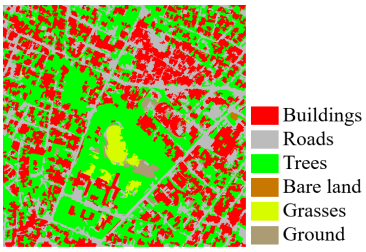


Fig. 9. Corresponding supervised classification result.

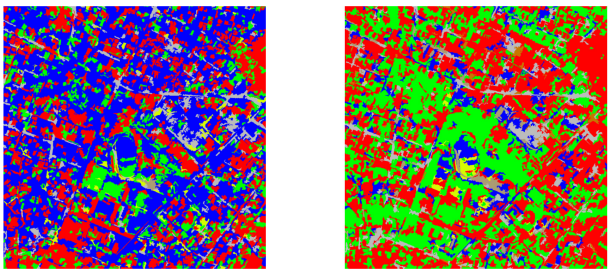


Fig. 10. (a) Result acquired by a nontransferred classifier. (b) Result acquired by MKJDM.

(Haiti, middle-east) scene in terms of the object shapes, color spectrums, and their distributions. Compared to the source domain, there are no water pixels the buildings are much denser in Haiti data set. Since the source and target domain share classes, in the Haiti data set, it is possible to classify nonwater pixels into water pixels. A notable improvement after DA is that pixels misclassified to water are significantly reduced in this data set. In addition, the target image resolution is lower than that in the source domain. We also adopt a family of RBF kernels whose bandwidths range between [0.025, 2] with a step-size of 0.025. The classification results acquired by MKJDM and a nontransferred classifier are shown in Fig. 10.

In this experiment, all six classes exist in Fig. 10(a). Therefore, all the instances in the source domain are reweighted in

TABLE III
RESULTS OF DIFFERENT DA METHODS FOR EXPERIMENT 2

DA Method	Overall Accuracy (OA)	Final Kappa Index (KIA)
Src-Tar (with DSM)	0.2990	0.1479
Tar-Tar (with DSM)	0.7998	0.6915
TCA (with DSM)	0.7132	0.5410
JDA (with DSM)	0.7410	0.5811
GFK (with DSM)	0.3459	0.2167
CORAL (with DSM)	0.3022	0.1520
TJM (with DSM)	0.7184	0.5401
JDM (with DSM)	0.7482	0.5955
MKJDM (with DSM)	**0.7652**	**0.6237**
Src-Tar (image only)	0.2902	0.1435
Tar-Tar (image only)	0.7574	0.6226
MKJDM (image only)	0.5306	0.4983

MKJDM. The comparative experiment results are shown in Table III.

All the DA methods are able to improve both OA and KIA of the classification result. Without transfer learning, the overall accuracy is only 29% in this experiment. MKJDM is still able to dramatically improve the classification result and significantly reduce the target domain areas where are misclassified to non-existing classes (water). The OA of MKJDM reaches 76.52% which is 46.62% higher than Src-Tar and 2.42% higher than the second-best DA method. At the same time, the KIA of MKJDM reaches 62.37%, 47.58% higher than Src-Tar and 4.26% higher than the second-best DA method. In addition, the OA is improved from 29.02% to 53.06% by our method in the case without DSM features. Using MKJDM, multimodel data set has a better transferability across different remote sensing data sets. This is mainly because height information is more stable across different data sets. Table III suggests that although the OA of MKJDM (76.52%) is only about 2% higher than JDM, the performance of MKJDM is already close to Tar-Tar (supervised method). In this case, it is hard for an unsupervised DA method to dramatically improve the OA. We conducted similar analyses as per

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TABLE IV
COMPARISON OF CLASS-WISE F1-SCORE FOR EXPERIMENT 2

Class	Src-Tar	Tar-Tar	JDM	MKJDM (without reweighting)	MKJDM
Buildings	0.4659	0.8047	0.6683	0.7251	0.7345
Roads	0.4665	0.7522	0.4649	0.4298	0.5070
Trees	0.3014	0.9461	0.8871	0.9195	0.9207
Bare land	0.0001	0.1939	0.0000	0.0000	0.1634
Grass	0.0455	0.6661	0.2327	0.1365	0.1527
Ground	0.4244	0.6454	0.4363	0.5478	0.6224

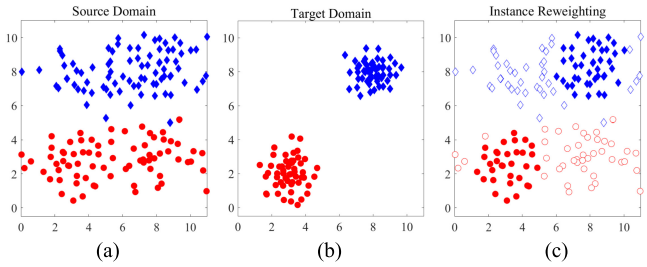


Fig. 11. (a) Instances in the source domain after aligning distribution. (b) Instances in the target domain after aligning distributions. (c) Instances in the source domain after aligning distribution and instance reweighting.

Experiment 1 on: 1) MKMMD with and without reweighting and 2) feature distribution before and after alignment, both suggest similar conclusions as in Experiment 1.

In general, reweighting instances improves classification performance in a stable way. In this experiment, we also fix the range of kernel bandwidth as [0, 2]. The step of bandwidths in the kernel family is not sensitive to MKJDM. Table IV shows the comparison of class-wise F1-score with different methods. Although the Argentina data set has seven classes and the Haiti data set has only six classes, MKJDM is also able to make the transferred data sets be closer in each class.

VI. CONCLUSION

In this article, we propose a novel unsupervised DA method, namely, MKJDM method. The proposed method is applied to the multi-modal remote sensing data set (Orthophoto + DSM) and the multi-spectral data set (with DSM). Based on our experiments, the proposed method achieved satisfactory results and has a promising potential to significantly reduce the label cost in classifying remote sensing images. To align the distributions of the source and target feature domains, our method considers formulating both marginal distribution and conditional distribution under an MK-MMD measure in a single optimization step. In addition, our proposed method reweights instances belonging to certain classes in the source domain to enhance the robustness of the feature distribution alignment. The solution of the formulated optimization delivers a transformation matrix which is used to map the feature sets of the source and target domains into RKHS can be computed, and the resulting feature distributions of the source and target domain are more statistically close. A simple statistical classifier using the transformed features for classification is able to acquire satisfactory results. As compared to other state-of-the-art DA methods, the proposed method obtained the best results in our experiments, and in the best case, has obtained 47% improvement as compared to a nontransferred classifier. Our experiment data sets of cross-continental regions suggest the practical potentials in using the unsupervised method for classification of remote sensing data through DA-based TL. In future works, we will consider including more stable features of the multi-modal to further improve the performance of the classification and reduce its computational complexity for the application to large-format remote sensing images.

APPENDIX

In this section, we introduce two DA methods that our proposed method is based on.

A. TCA

Through aligning marginal distributions between feature data sets from source and target domain, TCA generates better feature representations across domains. The kernel matrix $\mathbf{K}$ in (2) can be decomposed as $\mathbf{K} = (\mathbf{K}\mathbf{K}^{-1/2})(\mathbf{K}^{-1/2}\mathbf{K})$. Consider a matrix $\tilde{\mathbf{W}} \in \mathbf{R}^{(n_s+n_t)\times m}$ as a matrix that is able to transform the feature vectors to a m dimensional space. In general, $m \ll n_s + n_t$. The resulting kernel matrix is formulated as [7]

$$\tilde{K} = \left(\mathbf{K}\mathbf{K}^{-\frac{1}{2}}\tilde{\mathbf{W}}\right)\left(\tilde{\mathbf{W}}\mathbf{K}^{-\frac{1}{2}}\mathbf{K}\right) \tag{24}$$

where the rank of $\tilde{\mathbf{K}}$ is m, which means that the dimensions of the space corresponding to $\tilde{\mathbf{K}}$ is m. Then, the distance between the mapped feature sets $\mathbf{X}_S^*$ and $\mathbf{X}_T^*$ can be written as

$$\text{Dist}\left(\mathbf{X}_S^*, \mathbf{X}_T^*\right) = \text{tr}(\tilde{\mathbf{K}}\mathbf{M}). \tag{25}$$

Let $\mathbf{W} = \mathbf{K}^{-1/2}\tilde{\mathbf{W}} \in \mathbf{R}^{(n_s+n_t)\times m}$. By the definition of $\tilde{\mathbf{K}}$ in (3), the distance between $\mathbf{X}_S^*$ and $\mathbf{X}_T^*$ can be rewritten as

$$\text{Dist}\left(\mathbf{X}_S^*, \mathbf{X}_T^*\right) = \text{tr}(\mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{M}\mathbf{K}^{\mathrm{T}}\mathbf{W}) \tag{26}$$

where $\mathbf{W} \in \mathbf{R}^{(n_s+n_t)\times D}$ is the transformation matrix used to embed the features from their original space to RKHS. To minimize the criterion (26), a regularization term $\text{tr}(\mathbf{W}^{\mathrm{T}}\mathbf{W})$ is used to control the complexity of $\mathbf{W}$. And then, the kernel learning problem is then written as

$$\begin{aligned} &\min\ \text{tr}(\mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{M}\mathbf{K}^{\mathrm{T}}\mathbf{W}) + \lambda\text{tr}(\mathbf{W}^{\mathrm{T}}\mathbf{W}) \\ &\text{s.t. } \mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{M}\mathbf{K}^{\mathrm{T}}\mathbf{W} = \mathbf{I} \end{aligned} \tag{27}$$

where λ is a regularization parameter, $\mathbf{I} \in \mathbf{R}^{m\times m}$ and $\mathbf{I}_{n_s+n_t} \in \mathbf{R}^{(n_s+n_t)\times(n_s+n_t)}$ are the identity matrices, $\mathbf{H} = \mathbf{I}_{n_s+n_t} - (1/n_s + n_t)\mathbf{1}\mathbf{1}^{\mathrm{T}}$ is the centering matrix, and $1 \in \mathbf{R}^{n_s+n_t}$ is the column vector with all ones. The constraint $\mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{M}\mathbf{K}^{\mathrm{T}}\mathbf{W} = \mathbf{I}$ is to avoid the trivial solution ($\mathbf{W} = 0$). The readers may refer [7] for more details about TCA.

B. TJM

Inspired by TCA, TJM also adopts MMD as the nonparametric distance to measure the difference of feature distributions in RKHS, but TJM reweights the instances belonging to the source domain at the same time. The schematic of the reweighting instance is shown in Fig. 11.

Long *et al.* [12] proposed to impose the $\ell 2,1$-norm structured sparsity regularization on the transformation matrix W,

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which introduces row-sparsity to the transformation matrix. Since each row of matrix W corresponds to an instance, the row-sparsity is able to essentially facilitate adaptive instance reweighting. The instance reweighting regularizer is defined as

$$\|\mathbf{W}_s\|_{2,1} + \|\mathbf{W}_t\|_F^2 \tag{28}$$

where $\mathbf{W}_s = (1 : n_s, :)$ is the transformation matrix corresponding to the source instances, and $\mathbf{W}_t = \mathbf{W}(n_s + 1 : n_s + n_t, :)$ is the transformation matrix corresponding to the instances in the target domain. TJM only imposes $\ell_{2,1}$-norm regularization on source instances, since our aim is to reweight source instances by their relevance to the target instances. Then, the TJM optimization problem can be written as

$$\begin{aligned} &\min\ \operatorname{tr}(\mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{M}\mathbf{K}^{\mathrm{T}}\mathbf{W}) + \lambda\left(\|\mathbf{W}_s\|_{2,1} + \|\mathbf{W}_t\|_F^2\right) \\ &\text{s.t. } \mathbf{W}^{\mathrm{T}}\mathbf{K}\mathbf{H}\mathbf{K}^{\mathrm{T}}\mathbf{W} = \mathbf{I}. \end{aligned} \tag{29}$$

The readers may refer [12] for more details about TJM. However, neither TCA nor TJM aligns the conditional distribution of the source and target domains. Although the distance between their marginal distributions is minimized, the distance between their joint distribution may not be minimized. In MKJDM, we will use pseudo labels to align the marginal and conditional distributions at the same time.

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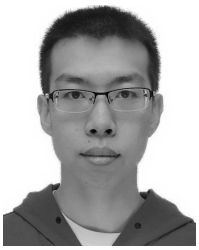
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